CS6570: Secure System Engineering

Assignment 2 : Return-to-libc (100 Marks)

You are provided with a dockerfile containing the binary similar to (but with some additional hidden functions) the one provided in the previous assignment. Your task is to do the following:

a) Similar to the Assignment 1, we have another function which you need to call by exploiting the vulnerability present in the welcome function. (Without the canary)
The function takes in a password as an argument, and a user is only provided access if the password matches.

Task 1 is to generate a payload, *payload_1* that can **call this function and pass the password check** to print the admin id. (10 marks)

b) The admin id printed in the hidden function is fixed, and for a certain specific id, the function prints some additional information. Use the ROP gadgets identified in the binary to generate a payload, *payload_2* that prints **both the specific id and the hidden print statement associated with the specific id**. (30 marks)

A **clean exit** after the execution of the 2 payloads carry **10 marks each**.
The report containing the explanation of the payload generation and gadgets identified and used (not just the address but the equivalent assembly instructions) would carry **40 marks**.

Submission guidelines:

You need to submit a zipped file named “<your_roll_no.>.zip” containing the following:

1) *payload_1*
2) *payload_2*
3) Scripts to generate the corresponding payloads
4) Report explaining the payload generation script and the gadgets used.